EXERCISE

A. MULTIPLE CHOICE QUESTIONS

Choose the correct answer:

i) The biological functions which perform gaseous exchange:
(a) Photosynthesis (b) Respiration
(c) Both a & b (d) Growth

ii) Plants do exchange of gases through:
(a) Roots (b) Stomata
(c) Stem (d) All of these

iii) Each stoma is formed by:
(a) one guard cell (b) two guards cells
(c) three guard cells (d) four guard cells

iv) Respiratory surface possesses following property:
(a) thin and wet (b) permeable
(c) very large (d) all of these

v) Inspiration involves:
(a) Contraction of intercostals muscles
(b) contraction of diaphragm
(c) Inward movement of ribs
(d) Both a & b

vi) Larynx is located on:
(a) Lungs (b) Trachea
(c) Bronchus (d) Bronchiole

vii) The respiratory surface of human is:
(a) Nostril (b) Bronchiole
(c) Alveoli (d) Trachea

viii) Increase in rate of breathing is due to the following:
(a) increase CO_2 in blood (b) Increase O_2 in blood
(c) decrease CO_2 in blood (d) decrease O_2 in blood

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ix) Which of the following disorder is associated with degeneration of alveoli?

(a) Bronchitis (b) Lung cancer
(c) Asthma (d) Emphysema

x) Which of the following disorder is associated with inflammation of air passage ways?

(a) Bronchitis (b) Lung cancer
(c) Asthma (d) Emphysema

B. SHORT QUESTIONS:

i) Why the stomata generally open during day-time?

ii) Which parts of the plant intake CO_2 and give out O_2, take in O_2 oxygen and give out CO_2 during day-time?

iii) Why do we have to breathe through nostrils rather than oral cavity?

iv) Differentiate between breathing, gaseous exchange and respiration.

v) Why do we deep breath during or immediately after exercise?

vi) What is "oxygen debt"?

vii) Distinguish between inspiration and expiration.

viii) What is lung cancer?

ix) How the asthma is characterized?

x) Name five animals which use their body surface for gaseous exchange.

C. EXTENSIVE RESPONSE QUESTIONS:

i) What measures would you take to avoid respiratory disorders?

ii) Discuss human respiratory system with the help of suitable illustrations.

iii) Prove with the help of experiment that CO_2 is released during respiration.

iv) Explain the process of ventilation in man.

v) Why smoking is dangerous? How it is related with respiratory disorders?

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EXERCISE

A. MULTIPLE CHOICE QUESTIONS

Choose the correct answer:

i) The internal condition of an organism is referred as:
(a) Homeostasis (b) Internal environment
(c) Internal metabolism (d) Feedback mechanism

ii) A set of metabolism reaction which maintain internal environment is:
(a) Positive feedback (b) Negative feedback
(c) Osmoregulation (d) Homeostasis

iii) Removal of extra liquid water is:
(a) Exudation (b) Guttation
(c) Respiration (d) Transpiration

iv) Plant grow near coastal area called:
(a) Xerophyte (b) Halophyte
(c) Epiphyte (d) Hygrophyte

v) Organ of human body which is considered on the largest organ is:
(a) Skin (b) Digestive tract
(c) Liver (d) Brain

vi) The maintenance of body temperature with in suitable limit is called
(a) Homeotherm (b) Thermoregulation
(c) Osmoregulation (d) Heterotherm

vii) The kidney is enclosed in a membrane called
(a) Pericardium (b) Peritoneum
(c) Pleural membrane (d) Plumule

viii) The network of blood capillaries present in the layer of skin:
(a) Epidermis (b) Dermis
(c) Hypodermis (d) Endodermis

ix) Selective reabsorption in nephron takes place at
(a) Glomerulus (b) Malpighian body
(c) Convulated tubules (d) Loop of Henle's

x) The hormone ADH release from
(a) Pituitary gland (b) Kidneys
(c) Liver (d) Lungs

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EXERCISE

A. MULTIPLE CHOICE QUESTIONS

Choose the correct answer:

i) The activity in relation to changes in environment is:
(a) Stimulus (b) Response
(c) Both a & b (d) None of these

ii) Stimulus is detected by:
(a) Receptor (b) Effector
(c) Nerve (d) All of these

iii) The type of coordination through electrochemical signals is:
(a) Nervous (b) Chemical
(c) Mechanical (d) All of these

iv) The chemicals released from one cell and carried to signal some distant cell through blood are:
(a) Neurotransmitters (b) Enzymes
© Hormones (d) all of these

v) The type of coordination exhibited by Plants:
(a) Nervous coordination (b) Chemical coordination
(c) Mechanical coordination (d) Both a & b

vi) The part of brain involved in reasoning is:
(a) Fore brain (b) Cerebrum
(c) Cortex (d) Frontal lobe

vii) The part of brain involved in balance and precision in movements is:
(a) Cerebrum (b) Cerebellum
(c) Thalamus (d) Medulla oblongata

viii) Vital functions for survival of animals are regulated by:
(a) CNS (b) PNS
(c) somatic sub-division (d) Autonomic sub-division

ix) The shortest path of reflex action consists of:
(a) 1 neuron (b) 2 neurons
(c) 3 neurons (d) many neurons

x) The type of lens in our eye is:
(a) Convex (b) concave
(c) Both a & b (d) None of these

xi) The automatic process of altering focus to get sharper image of near object is:
(a) Vision (b) Accommodation
(c) Focus (d) All of these

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xii) The vitamin necessary for proper vision is:
(a) Vitamin A (b) Vitamin B
(c) Vitamin C (d) Vitamin D

xiii) A colour blind person cannot see:
(a) Anything (b) Red
(c) White (d) Black

xiv) The book "Kitab-ul-manazir" was written by:
(a) Jabir bin Hayan (b) Ibn-al-Haitham
(c) Ali Ibn-Isa (d) Bu-Ali Sina

xv) Sensory hair-cells are present in:
(a) Retina (b) Cochlea
(c) Skin (d) Nose

xvi) The gonads are the target organ for:
(a) FSH (b) LH
(c) Both a & b (d) None of these

B. SHORT QUESTIONS:

i) What do you mean by feedback system?
ii) Why the nervous coordination is faster than chemical coordination?
iii) Which of the two coordination types is better and why?
iv) How reflex action works by a reflex-arc?
v) Why driving license is not issued to a color blind person?
vi) Why thyroid gland swell up and give the name of the disease?
vii) What is the role of Islet's of Langerhans cells?
viii) What is "emergency hormone" and why it is named so?
ix) What are the possible reasons of paralysis?
x) What is Epilepsy?
xi) Differentiate between Chemical and Nervous Coordination.

C. EXTENSIVE RESPONSE QUESTIONS:

i) Discuss the detailed structure of human brain?
ii) Write a note on human eye or ear.
iii) Why does the intelligence level of persons differ from each other, even though the components of brain are same?
iv) What is endocrine system? Discuss the gland involved in regulation of blood glucose and how?
v) What is diabetes mellitus? Explain some ways of its management.

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EXERCISE

A. MULTIPLE CHOICE QUESTIONS

Choose the correct answer:

i) Irritation caused by stimulus is the main cause of
(a) Tropism (b) Movement
(c) Locomotion (d) Arthritus

ii) The frame work which gives shape to any structure called
(a) Architecture (b) Bone
(c) Cartilage (d) Skeleton

iii) The cartilage are made up of cells called
(a) Osteoclast (b) Osteocytis
(c) Chondrocytes (d) Chaonocytes

iv) The head of femur attached with
(a) Pelvic girdles (b) Pectoral girdle
(c) Scapula (d) Acetabulum of pelvic gridles

v) The large muscles of arm is
(a) Bicep (b) Tricep
(c) Tetracep (d) Pentacep

vi) The band of tough, fibrous, connective tissue which are attached to bone at joints called
(a) Ligament (b) Tendon
(c) Bicep (d) Tricep

vii) Softening and weakening of bone in children due to deficiency of vitamin D called
(a) Osteoporesis (b) Osteoarthritis
(c) Rickets (d) Rheumatic fever

viii) The muscle which is responsible to straighten the limb is
(a) Ligament (b) Skeleton muscle
(c) Flexor (d) Extensor

ix) The pair of muscle where both work in opposite direction
(a) Antagonist (b) Cardiac
(c) Smooth (d) Abductor

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- Spermetogenesis is the formation of sperm, where dogenesis is the formation of ovum.
- Male and female reproductive organs are
  - i) Gonades to produce gamets
  - ii) Ducts; gamets collecting tubes
  - iii) Genitals; gamets donating or receiving organs.
- Glands of male are prostrats, cowper's and seminal vesicle.
- Glands of female are ovaries.
- Fertilization is the fusion of male and female gametes form diploid zygote (2N)
- There are two types of fertilization i-e external and internal fertilization.
- Population planning is a policy to limit the growth in number of population.
- Sexually transmitted disease are those which are passed from one person to another through genital organ and genital fluid.

EXERCISE

A. MULTIPLE CHOICE QUESTIONS

Choose the correct answer:

i) The process which is essential for continuing and survival of species is

(a) Digestion (b) Respiration
(c)Reproduction (e)Excretion

ii) The type of reproduction which is necessary for evolution is

(a) Vegetative propagation (b) Fragmentation
(c) Sexual reproduction (d) clonning

iii) The unicellular structure, responsible for asexual reproduction without fusion is

(a) Pores (b) Spores
(c) Gametes (d) Pollen grains

iv) The example of stem which run horizontally on surface of soil to produce vegetatively

(a) Mint (b) Ginger
(c) Onion (d) Bryophyllum

v) Plant stem that arise from buds on the base of parent plants are

(a) Bulb (b) Rhizome
(c) Sucker (d) Runner

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vi) The type of seed production without fusion of male and female gametes is
(a) Parthenocarpy (b) Apormixes
(c) Grafting (d) Scion

vii) The female gametophyte of anglosparmic plant is
(a) Embryo sec (b) Ovule
(c) Ovary (d) Carpel

viii) The 3N zygote is angiosperm develop into
(a) Seed coat (b) Cotyledon
(c) Embryo (d) Endosperm

ix) The male gonades in rabbit are
(a) Testis (b) Ovaries
(c) Scrotal sec (d) Vasdeferens

x) The female gametes are fertilized in the rear end of
(a) Oviduct (b) Follopin tube
(c) Ovaries (d) Both a and b

B. SHORT QUESTIONS:

i) Distinguish between asexual and sexual reproduction, Epigial and Hypogial Germination.
ii) Draw neat and labeled diagram of T.S of Angiospermic flower.
iii) How a new plant develops with an already growing plant?
iv) How leaves develop into new plants?
v) Draw neat and labelled diagram of Ovule.
vi) Draw neat and labelled diagram of male gametophyte of angiospermic plant?
vii) List out the male reproductive organs of rabbit with glands.
viii) What are STD's?
ix) Why population control is considered important for prosperous society?
x) Draw life cycle of Angiospermic plant.

C. EXTENSIVE RESPONSE QUESTIONS:

i) Describe natural asexual reproduction in plants.
ii) What is pollination? give adaptive characters of wind and insect pollinated flowers.
iii) Describe different types of asexual reproduction in animals.
iv) Describe the process of Spermetogenesis.
v) What is germination? Give the condition of germination also describe the methods of different germination.

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- The phenomenon of inheritance where both alleles of a trait expressed in heterozygous condition and their expression get blended to produce new phenotype called **incomplete Dominance** or **Partial Dominance**.
- The differences in characters such as height, colour, etc. among individuals of same species are called Variations. Variation may be caused either by the effect of environment or by the changes in the genetic material.



A. MULTIPLE CHOICE QUESTIONS

Choose the correct answer:

i) Branch of biology deals with the study of heredity and variation is called:

(a) Inheritance (b) Heredity
(c) Genetics (d) Evolution

ii) The way by which gene transmits characters from parents to offspring is

(a) Genetics (b) Inheritance
(c) Heredity (d) Allele

iii) Two similar chromosomes in a cell which are similar in shape, size and position of centromere called

(a) Chromatids (b) Arms
(c) Homologous (d) Homology

iv) The chemical material of a chromosome is called

(a) Chromatin (b) Chromeres
(c) Chromonema (d) Chromatid

v) The outer part of DNA helix made up of sugar and phosphate is called

(a) Nucleoprotein (b) upright
(c) Rungs (d) Phosphoester

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vi) The small segment of DNA which has information to code one protein is called

(a) Nucleotide (b) Polynucleotide
(c) Gene (d) Exon

vii) The exact duplication of DNA is called

(a) Duplication (b) Replication
(c) Transcription (d) Translation

viii) If both the parents donate same factors of a character is called

(a) Homologous (b) Heterologous
(c) Homozygous (d) Heterozygous

ix) The Genetic constitution of a trait is called

(a) Genotype (b) Phenotype
(c) Genome (d) Phenyl

B. SHORT QUESTIONS:

i) What is gene and how it works?

ii) Define following terms

(a) Homologous Chromosome (b)Heterologous Chromosome
(c) Dominant allele (d) Allele
(e) Recessive allele (f) Homozygous
(g) Heterozygous

iii) Which of the phenomena of inheritance is there where both factor express in heterozygous condition.

iv) List out the factors which are involved in theory of natural selection.

v) Relate artificial selection with natural selection.

C. EXTENSIVE RESPONSE QUESTIONS:

i) Describe the Waston and Crick model of DNA.

ii) How gene express describe in detail.

iii) State and explain the law of segregation of Mendel's.

iv) Explain the inheritance of dihybrid cross by law.

v) Explain the Darwin of theory of natural selection.

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EXERCISE

A. MULTIPLE CHOICE QUESTIONS

Choose the correct answer:

i) The life sustaining envelop of earth is

(a) Biomass (b) Biomes
(c) Biosphere (d) Atmosphere

ii) The group of organisms belong to the same species live in a particular area called

(a) Community (b) Species
(c) Genes (d) Population

iii) An area where community interacts with non-living environment is called

(a) Community (b) Ecology
(c) Ecosystem (d) Biome

iv) Any biological region recognized by its climate or vegetation is called

(a) Ecosystem (b) Biomes
(c) Biosphere (d) Biomass

v) The transfer of food material from producers through some organisms with repeated eating and being eaten is called

(a) Food pyramid (b) Food chain
(c) Food web (d) Ecological pyramid

vi) The elements in ecosystem recycle through organisms with environment is called

(a) Food chain (b) Food web
(c) Chemical cycle (d) Biogeochemical

vii) Process where nitrogenous compound of living organism convert into nitrates is called

(a) Ammonification (b) Nitrification
(c) Denitrification (d) Deamination

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viii) The cold war between the organisms of a community occupying the same habitat is called

(a) Predation (b) Competition
(c) Mutualism (d) Commensalism

ix) The association between two different types of organism which get benefit to each other, cannot live without each other is called

(a) Parasitism (b) Commensalism
(c) Mutualism (d) Predation

x) The amount of solid waste or concentration of gases other than oxygen increase in atmosphere is called

(a) Air pollution (b) Ozone depletion
(c) Acid rain (d) Greenhouse effect

B. SHORT QUESTIONS:

i) What is the effect of light on ecosystem?
ii) Which one is the first trophic level of ecosystem and how?
iii) What do we mean by nitrogen fixation and how it occurs in an ecosystem?
iv) What is interaction?
v) What is pyramid of number?
vi) What is greenhouse effect?
vii) What is algal bloom and how it destroy the life of an aquatic ecosystem?
viii) What measure can be taken to control water pollution?
ix) Write down the name of some endangered mammals of Pakistan.
x) Draw the diagram of nitrogen cycle.

C. EXTENSIVE RESPONSE QUESTIONS:

i) Describe biotic factors of an ecosystem.
ii) Describe nitrogen cycle as biogeochemical cycle in detail.
iii) What is pollution? Describe different types of hazardous effects caused by air pollution.

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- Single-Cell Protein (SCP) is a term represents microbial cells (primary) grown in mass culture and harvested for use as protein sources in foods or animal feeds.
- Single Cell Protein (SCP) is not a pure protein but refers to whole cells of bacteria, yeasts, filamentous fungi or algae.
- Many types of animal feeds contain single cell proteins. 60-80% dry cell weight; contains nucleic acids, fats, CHO, vitamins, and minerals rich in essential amino acids (Lys-Met).



A. MULTIPLE CHOICE QUESTIONS

Choose the correct answer:

i) The artificial manipulation, medication and recombination of DNA

(a) Genetic engineering (b) Biotechnology
(c) Molecular biology (d) Genetic

ii) The earlier biotechnologist were

(a) Biologist (b) Agriculturist
(c) Genetist (d) Farmers

iii) The complete graph of human genome was studied by

(a) PCR (b) HGP
(c) Medicine (d) soma-clonal

iv) Alcohol and antibiotics on large scale production by organism is an area of

(a) Environmental biotechnology (b) Fermentation
(c) Biotechnology in industry (d) Medicinal biotechnology

v) Most of the living things use O_2 to produce

(a) ATP (b) Alcohol
(c) Organic acid (d) Ecological pyramid

vi) In acidic formulation lactic acid produced from

(a) Pyruvic acid (b) Acetic acid
(c) Citric acid (d) Glyceric acid

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vii) The bread dough rises during alcoholic fermentation is due to

(a) Methyl alcohol (b) CO_2

(c) Ethyl alcohol (d) H_2O

viii) The container use to grow bacteria on large scale are called

(a) Chillers (b) Sterilizers

(c) Fermenters (d) Ferments

ix) Naturally occurring enzyme used as a defense chemical by bacteria

(a) Defense protein (b) Restriction enzyme

(c) Hydrolytic enzyme (d) ligase enzyme

x) Extra circular DNA which use as vector of gene is

(a) Genome (b) Plasmid

(c) Pilli (d) Ligase

B. ANSWER THE FOLLOWING QUESTIONS:

i) What kind of enzymes allows scientists to cut and paste pieces of DNA together to form recombinant DNA?

ii) Explain how making human tissue plasminogen activator (t-PA) in Chinese hamster ovary (CHO) cells is an example of genetic engineering.

iii) How do organisms obtain energy? How does fermentation work?

iv) What types of microorganisms cause fermentation to occur?

v) What food and non-food products are created by fermentation?

vi) What are some advantages and disadvantages of fermentation in food processing?

vii) What factors can affect the fermentation process?

viii) Define biotechnology?

ix) What is the classical biotechnology?

x) What is your justification in considering biotechnology an old technology?

xi) Describe the contribution of Louis Pasteur toward development of classical biotechnology.

xii) What are the different types of fermentation?

xiii) What is lactic acid bacillus? Explain its role in the formation of curd.

xiv) Name some of the fermented food products of Pakistan.

xv) Name four recombinant DNA products available on the market.

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SUMMARY

- Pharmacology is the branch of biomedical science which is concerned with the use, effect and mode of actions of drugs.
- Drug is a chemical substance which is used to treat, cure, prevent a disease and used for artificial pleasure.
- Drugs can be derived from plants and animals.
- There are two categories of drugs
  i. Pharmaceutical or medicinal drug **ii.** Addictive drugs or narcotics
- Medicinal drugs are derived from micro organisms, mineral, plants, animal or synthetic.
- Medicinal drugs may be pain killer, antibiotics, vaccine, sedative, antiseptics.
- Addictive drugs may be sedative, narcotics (Heroine, Morphine, Marijuana).
- Antibiotics are bacteriostatic or bactericidal.
- Bacteriostatic mean inhibit growth of bacteria.
- Bactiocidial means kill the bacteria.
- Antibiotics work against wide range of Gram+ve and Gram-ve bacteria called broad spectrum antibiotics.
- A vaccine is a biological preparation that improves immunity to a particular disease.
- Immunization is the process whereby a person is made resistant to an infectious disease.

EXERCISE

A. MULTIPLE CHOICE QUESTIONS

Choose the correct answer:

i) Who is called father of antiseptic?

(a) Alexandar fleming (b) Edward Jenner
(c) Lister (d) Oswald Schmiedeberg

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ii) Drugs for treatment of rheumatoid arthritis can be obtained from:

(a) Animals (b) Minerals
(c) Plants (d) Microorganisms

iii) Drugs that slow normal brain functioning are categorized as:

(a) Narcotics (b) Hallucinogen
(c) Marijuana (d) Sedatives

iv) Vaccination can be administered :

(a) After infection (b) Before infection
(c) During infection (d) All are Correct

v) The substance which inhibit the growth of bacteria can be considered as:

(a) Vaccine (b) Bactericidal
(c) Bacteriostatic (d) Antibiotics

vi) Haris is addicted to a drug , which left the following effect on Haris :

i. Blurred vision
ii. Making unseen faces in imagination
iii. Euphoria

Identify the drug, to which Haris is addicted?

(a) Narcotics (b) Hallucinogen
(c) Antibiotics (d) Antiseptic

vii) Which one is not the effect of misuse of antibiotics?

(a) Diarrhea (b) Immunization
(c) Stomach upset (d) Antibiotic resistance

B. SHORT QUESTIONS:

i) Why antibiotics are not effective against viral infection?
ii) Why the sedative is used for?
iii) Why addiction is considered as harmful condition?
iv) How drugs (medicine) can be taken from natural source?
v) Is it possible to get drugs from animals name some of them?
vi) Do we have any harm of antibiotics? If, so mention them.
vii) How vaccine work against pathogen? Explain the process with the help of diagram.

C. EXTENSIVE RESPONSE QUESTIONS:

i) How bacteria produces antibiotic resistance?
ii) Describe the mode of vaccination.

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